PYTHON-BASED HYDROPOWER GENERATION AND PERFORMANCE ANALYSIS

Hydroelectric Energy Production Modeling, Hydraulic Losses and Design Optimization

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Environment: Python / Google Colab

Libraries: NumPy, Pandas, Matplotlib; HydroGenerate investigated

Draft status: Numerical HydroGenerate comparison remains to be verified.

Abstract

This report presents an educational Python-based simulation of hydroelectric energy generation. The model evaluates monthly inflow, ecological flow release, turbine flow capacity, penstock diameter, hydraulic losses and overall efficiency. Under the stated illustrative assumptions, the baseline case produces 102055.61 MWh (102.056 GWh) per year. The highest-energy option in the tested design grid produces 110.426 GWh/year. These are simulation results, not measured data from an operating plant.

1. Introduction

Hydropower generation depends on available flow, hydraulic head and energy-conversion efficiency. Practical performance is also affected by ecological flow requirements, turbine capacity limits and hydraulic losses. This project builds a transparent introductory model in Python and explores these relationships.

2. Scope and assumptions

All monthly inflow values are assumed teaching data, not measurements from Kav■ak Bendi HEPP or another real facility. A 10% ecological release is deducted from monthly inflow and the remaining flow is limited by turbine capacity.

Parameter	Assumption
Monthly inflow (Jan–Dec)	12, 14, 18, 25, 30, 28, 22, 16, 12, 10, 9, 11 m³/s
Gross head / overall efficiency	100 m / 85%
Ecological release / max turbine flow	10% / 20 m³/s
Penstock length / diameter	1,000 m / 3.0 m
Darcy friction factor / minor-loss K	0.02 / 2.0
Water density / gravity	1,000 kg/m³ / 9.81 m/s²

3. Methodology

For each month, turbine flow is the lesser of the flow remaining after ecological release and the maximum turbine flow. Penstock velocity is calculated from flow and pipe area. Darcy–Weisbach friction loss and a lumped minor loss are subtracted from gross head to obtain net head. Electrical power is calculated from overall efficiency, water density, gravity, turbine flow and net head. Monthly energy equals power multiplied by the number of operating hours in the month.

Core equations

$$Q_t = \min[Q_{in} \times (1 - r_{eco}), Q_{max}]$$
$$A = \pi D^2/4 \; ; \; v = Q_t/A$$
$$h_f = f(L/D)(v^2/2g) \; ; \; h_m = K(v^2/2g)$$
$$H_{net} = H_{gross} - h_f - h_m$$
$$P = \eta \rho g Q_t H_{net} \; ; \; E_{month} = P \times 24 \times N_{days}$$

4. Baseline monthly results

The baseline configuration gives 102055.61 MWh/year (102.056 GWh/year). Annual average power is 11.650 MW. The time-weighted mean hydraulic loss is 1.991 m and the time-weighted mean net head is 98.009 m.

Month	Inflow	Turbine Q	Net head	Power	Energy
January	12.0	10.8	98.97	8,913	6,631
February	14.0	12.6	98.60	10,359	6,961
March	18.0	16.2	97.68	13,195	9,817
April	25.0	20.0	96.46	16,087	11,583
May	30.0	20.0	96.46	16,087	11,969
June	28.0	20.0	96.46	16,087	11,583
July	22.0	19.8	96.53	15,938	11,858
August	16.0	14.4	98.17	11,787	8,770

Month	Inflow	Turbine Q	Net head	Power	Energy
September	12.0	10.8	98.97	8,913	6,417
October	10.0	9.0	99.28	7,451	5,543
November	9.0	8.1	99.42	6,715	4,835
December	11.0	9.9	99.13	8,184	6,089

Units: flow in m³/s, head in m, power in kW, energy in MWh.

5. Design parameter comparison

A discrete sweep compared penstock diameters of 2.5–4.5 m and maximum turbine flows of 12, 16, 20 and 24 m³/s. The highest annual energy in this limited grid is 110.426 GWh/year at a diameter of 4.5 m and maximum turbine flow of 24 m³/s. This is an energy-only result, not an economic optimum.

D (m)	12 m³/s	16 m³/s	20 m³/s	24 m³/s
2.5	78.52	90.86	98.62	103.07
3.0	79.78	93.13	102.06	107.63
3.5	80.25	93.97	103.34	109.33
4.0	80.45	94.34	103.89	110.07
4.5	80.55	94.52	104.16	110.43

The baseline 3.0 m / 20 m³/s case yields 102.056 GWh/year. The best option in this limited grid yields 8.20% more energy. Larger waterways and turbines may require higher capital investment and may face civil, hydraulic and environmental constraints.

6. Interpretation

Higher inflow generally increases generation until the turbine flow limit is reached. Ecological release reduces the water available for generation but represents an important environmental operating requirement. Increasing penstock diameter reduces velocity and modeled hydraulic losses for a given flow. A larger turbine capacity captures more available flow during high-inflow months. The highest-energy design in a limited grid is not automatically the least-cost or technically feasible design.

7. HydroGenerate comparison

HydroGenerate was installed in the Google Colab environment and the calculate_hp_potential function interface was inspected. A numerical comparison is not included because the exact returned output values, column definitions, units and annual aggregation have not been independently verified in the available project record. Complete this section after confirming matching flow time steps, ecological-flow treatment, head-loss assumptions, turbine type and efficiency basis.

8. Limitations and further work

The flow profile is illustrative; monthly averages conceal daily variability. Overall efficiency and Darcy friction factor are fixed. Minor losses are represented by one assumed coefficient. The model excludes water hammer, cavitation, sediment, outages, detailed turbine curves and grid constraints. Economic optimization requires capital and operating costs, discount rate, project life and electricity-price assumptions. Future work should use measured flow data, validate against a documented engineering tool, and perform sensitivity and uncertainty analysis.

9. Conclusion

The project demonstrates how a transparent Python model can explore hydropower generation and selected design parameters. Under the assumed baseline conditions, annual energy is approximately 102.056 GWh. In the tested design grid, the maximum is 110.426 GWh/year. The results are an educational simulation, not a feasibility study or prediction for a real plant. Site-specific conclusions require measured data and further validation.

References

Idaho National Laboratory, HydroGenerate documentation: https://idaholabresearch.github.io/HydroGenerate/

HydroGenerate package, Python Package Index: https://pypi.org/project/HydroGenerate/

All numerical assumptions and model equations are stated in this report.